

Forensics Lab: Using Properties to Identify Materials

Background: Forensic chemists test the physical and chemical properties of materials at a crime scene as part of evidence gathering. They also do similar tests on the materials found on a suspect's skin or clothing. These materials are often complex mixtures, such as soil, which contain many substances.

In this lab, you will compare the properties of three known substances with two samples of "evidence". These samples represent evidence from a crime scene and evidence from a suspect's shoe. Although your materials and equipment are less complex than those used by forensic chemists, your overall method will be similar to the methods they use.

Scenario: A home invasion robbery has just occurred. Crime scene investigators are gathering evidence from the kitchen area where the presence of white powders can be seen on the floor and counter tops. Police have a suspect in custody and they found traces of white powder on the bottom of his shoe. Working in a criminal forensics lab, it is your job to perform tests on the samples gathered and present it to the prosecutor.

Problem: Can the properties of substances be used to tell them apart and also identify an unknown?

Materials:

- well plate
- cornstarch
- baking powder
- baking soda
- crime scene sample
- suspect shoe sample
- water
- vinegar
- iodine solution *CAUTION: Iodine solution is corrosive. It stains skin and clothing. Rinse spills with water!*

Procedures:

Part A. Testing Properties of Known Substances

1. Put a small amount of cornstarch into 3 wells, baking soda into 3 wells, and baking powder into 3 wells (9 wells total). Remember where you put them by drawing a labelled diagram on the back of this handout.
2. Observe the samples and record any noteworthy physical observations in the table below.
3. Add a few drops of **water** to the first cornstarch, baking soda, and baking powder samples. Record observations and any changes that occur using the table on the next page.
4. Add a few drops of **vinegar** to the second cornstarch, baking soda, and baking powder samples. Record observations and any changes that occur using the table on the next page.
5. Add 1 drop of **iodine solution** to the third cornstarch, baking soda, baking powder samples. Record observations and any changes that occur using the table on the next page.

Wash and dry the well plate to do the next part.

Part B. Testing Properties of Crime Scene and Shoe Samples

1. Put a small amount of the crime scene sample into 3 wells, and then a small sample of the shoe samples into 3 wells (6 wells total). Remember where you put it by drawing a diagram again if needed.
2. Observe the samples and record any noteworthy physical observations in the table below.
3. Perform the same tests on each sample (as in Part A) by adding the same amount of tap water, vinegar, and iodine solution to each well. Record observations and any changes that occur using the table on the next page.

Rinse all materials off the well plate and flush them down the drain with plenty of water. Dry plate and return to tray.

Data Table: Observations of chemical changes during white powder analysis

	Powder alone	Powder with water	Powder with vinegar	Powder with iodine
Cornstarch				
Baking soda				
Baking powder				
Crime scene sample				
Shoe sample				

Analysis:

1. Give one example of a physical property and one example of a chemical property from this lab.

2. What is the difference between a physical property and a chemical property?

3. Solubility is the ability of a substance to dissolve (mix) in water. Describe the difference in solubility of cornstarch, baking soda, and baking powder. (I.e. Which one is most soluble, least soluble, moderately soluble?)

4. Describe the solubility of the crime scene sample as compared to the shoe sample.

5. What did you see that shows a chemical reaction with the vinegar? Describe the difference in reactivity of cornstarch, baking soda, and baking powder to vinegar. (i.e. Which is has the strongest reaction, weakest reaction, moderate reaction?)

6. Describe the reactivity to vinegar of the crime scene sample as compared to the shoe sample.

7. What did you see that shows a chemical reaction with the iodine solution? Describe the difference in reactivity of cornstarch, baking soda, and baking powder to iodine solution. (i.e. Which is has the strongest reaction, weakest reaction, moderate reaction?)

8. Describe the reactivity to iodine solution of the crime scene sample as compared to the shoe sample.

9. Are the samples from the suspect's shoe and from the crime scene identical? Justify your answer.

10. The prosecutor in this case wants to use the evidence from this lab to say that the suspect is guilty of the crime. Does she have a strong case? Explain your reasoning.

11. List 3 other properties (physical and/or chemical) that could be used to test the samples in this lab.

CRIME SCENE INVESTIGATION

S T F B F Z O W W E A O F R A S L A C I M E H C
L E X G A L Y U C O H M J X P X E Y D G J W E S
D O U B L E H E L I X M A N D H C U O T L M X N
T A X B Z U F C W H N T T X V S H O E P R I N T
X G B Y F F A T T Z F O Y X J Y V L K R B R R R
N W G G Y M G H C V Z J C J J L B T F A F B R E
I Q F P E S R O Y E B B R N F S I W I G X I H U
J U U R B R Q N K B V H S C C Z Y H N P B J S G
Q W A S G N U L G B C J N D Q Z X C G O R I N N
U L Y Q A D O X X R D B Z R Z Z Q N E U T E C D
Z M V R W F P E A A B N D E U J Q D R O G I R W
E J R E D W O P K C A L B R B C S W P I G F Z M
Z E I A X E C N E C S E R O U L F R R C B P S G
S I R E T A W D E L L I T S I D B S I R O U Y K
Z P S B A W S M N W E H O N B N I F N Y N B N U
O D A E N L D S F O D H F V L T O J T R Z D Q I
I V Y T F G M S B E D H G O O A M U S O L U W Q
M E G U T M G A D M D U O G C J E L R T O T Z V
S L N P G E G L B N T P B P K N T J E C Y A V R
O O T F Q H R G A L I G W Z C U R L G E U E T J
T C S Y L P H Y C R G V V M P F I K N J L D G G
G I E W C O O P F O F V W P Y N C T P A I I A O
Q T N T P V F S C H B G D Q B Q S M R R C E A N
T Y K L E A M D G W M D I A B G T J M T J V R D

ARCH	BIOMETRICS	BLACK POWDER	CAMERA
CHEMICALS	DISTILLED WATER	DOUBLE HELIX	FINGERPRINTS
FLUORESCENCE	LOOP	SHOEPRINT	SPATTER
SPYGLASS	SWABS	TOUCH DNA	TRAJECTORY
VELOCITY	WHORL		